

Vol 4 — General Relativity & Cosmology from D_IV⁵

Legacy chapter (May 2026), corrected in place 2026-09-11 (Keeper, K1892) at the sentences the presentation scan flagged; the remaining tiers are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23) — chapter content TO BUILD Lead: Lyra (primary, signature BST domain) + Elie (computational verification) + Grace (catalog backbone) + Keeper (audit)

Reader target

Graduate physics with GR / cosmology specialization. Reads Vol 0 (Substrate Foundation) + Vol 1 (QFT from D_IV⁵) as prerequisite; ideally also Vol 3 (Nuclear & Atomic for BBN connection). Vol 4 derives general relativity + Λ CDM + cosmological structure from substrate.

Prerequisites

- Vol 0 Substrate Foundation (D_IV⁵ + five integers + Bergman curvature + operator zoo)
- Vol 1 QFT from D_IV⁵ (Bergman Hilbert space + Casimir + Higgs + gauge fields)
- Vol 3 Nuclear & Atomic (optional but useful for Ch 8 BBN abundances)
- Standard graduate GR background (MTW / Wald / Carroll level) + observational cosmology familiarity

BST integer focus

GR & Cosmology is the signature BST domain — substrate-derived gravity + cosmological constant Λ + Hubble + CMB + BBN all from the five integers and substrate dynamics. Casey's directive Wednesday EOD: "D_IV⁵ as emergent TOE" — Vol 4 is where this lands operationally.

Key integers active in Vol 4: - All 5 integers + $N_{\text{max}} = 137$: cosmological observables span the integer web - C_2, g, rank : $\Lambda = g \cdot \exp(-C_2(g^2 - \text{rank})) = 7e^{-282}$ (T1485, 0.076 dex) - N_c, n_C : $\Omega_\Lambda = (N_c + 2n_C)/(N_c^2 + 2n_C) = 13/19$ (0.07 σ) - $C_2, n_C^2 + 2n_C$: $\Omega_m = C_2/(N_c^2 + 2n_C) = 6/19$ (0.07 σ) - $5/137 = n_C/N_{\text{max}}$: spectral index $n_s = 1 - n_C/N_{\text{max}} = 0.9635$ (0.3 σ) - DM/baryon: $(3n_C + 1)/N_c = 16/3$ (0.58%)

Chapter outline (10-12 chapters, TO BUILD)

Ch	Title	Existing BST coverage	Key theorems / toys
1	Newton's G from Bergman Curva- ture — P-3 BST-GR founda- tion	~80% (Paper #122 + AB-10)	$G = \hbar c (6\pi^5)^2 \alpha^{24} / m_e^2$, 12 = $2C_2$ Bergman round trips (0.07%)
2	Gravity as Cumu- lative Eigen- tone Effect — substrate dynam- ics → metric	~60% (AB-11 + T2418)	T2418 $\Lambda \leftrightarrow$ Casimir + AB-11 gravity-from-eigentones
3	BST-SR / BST-GR Bound- ary — when does GR emerge from sub- strate?	~50% (AB-12)	substrate at Planck → GR at macroscopic; Koons tick + Casimir crossover
4	The Cosmo- logical Constant Λ from Substrate — $\Lambda =$ $g \cdot \exp(-C_2(g^2 - \text{rank}))$	~85% (T1485 + AB-9)	T1485 (0.076 dex); Wednesday T2418 $\Lambda \leftrightarrow$ Casimir unification

Ch	Title	Existing BST coverage	Key theorems / toys
5	Hubble Constant — Four Routes (A/B/C/D) — tension resolu- tion via BST	~80% (4 independent routes, all at ~67-68 km/s/Mpc)	Route A: from η via Λ CDM; Route B: $\sqrt{(19\Lambda/39)}$; Route C: full CAMB Toy 677 at 0.1%; Route D: $c\sqrt{(19\Lambda/39)}$ Toy 903
6	CMB Structure — n_s , Ω_m , Ω_Λ , T_{CMB} from BST integers	~90% (Tasks #83 + #100 + #101)	$n_s = 1 - 5/137$ (0.3 σ); Ω_Λ = 13/19 (0.07 σ); T_{CMB} = 2.737 K Toy 904 (0.43%)
7	Inflation Param- eters — r/s , n_t , α_s running	~75% (Task #85)	$r \approx 0$ ($T_c \ll m_{\text{Pl}}$); $\alpha_s =$ $-(n_s-1)^2 = -25/18769$ (0.5 σ)
8	BBN — Element Abun- dances + Li-7 Problem — pri- mordial nucle- osynthe- sis from BST	~80% (Task #84)	$T_c = N_{\text{max}} \times 20/21 =$ 0.487 MeV (0.018%); ${}^7\text{Li}/\text{H}$ = $\sim 1.7 \times 10^{-10}$ (7%) via $\Delta g=7$ ($7 = p + q$, not the genus — corrected 2026-09-11)

Ch	Title	Existing BST coverage	Key theorems / toys
9	Cosmological Cycle Hypothesis + Interstasis — D_IV ⁵ Rigidity Principle observational signatures	30% (Task #267 ACTIVATED Friday + Casey-named #7)	T2467 Rigidity-as-Singleton + T2468 Rigidity-as-Unification (multiverse closure); annealing toward Λ -saturated state
10	Dark Energy + Dark Matter — Shannon channel framing; DM ratio 16/3	~85% (Task #139 + Dark matter ratio entry)	DM/ $\Omega_b = (3n_C+1)/N_c = 16/3$ (0.58%); galaxy rotation curves Shannon S/N
11	Gravitational Waves — NANOGrav prediction + black holes as eigentone	70% (Tasks #103 + #136 + #137)	GW peak frequency 6.4 nHz (NANOGrav); spectral index $\gamma = 13/5+1 = 3.60$; black holes as eigentone configurations
12	Observational Reanalysis Program (SP-27) — recalculate observations vs BST integer predictions	30% (SP-27 pending Casey scoping)	reanalysis methodology + falsifier specification

Approximately 70% existing BST coverage maps to specific chapters above. 30% to develop — primarily Ch 9 (cosmological cycle / Interstasis observable signatures), Ch 12 (SP-27 program scoping).

Cross-volume reference map

Inputs from Vol 0: - D_{IV}^5 + five integers + Bergman curvature - T2418 $\Lambda \leftrightarrow$ Casimir vacuum unification - T2417 cosmological cycle hypothesis (3-scale substrate operation) - T2467/T2468 D_{IV}^5 Rigidity Principle (Casey-named #7) — multiverse closure structurally

Inputs from Vol 1: - Bergman Hilbert space $H^2(D_{IV}^5) \rightarrow$ Bergman curvature $\rightarrow$ Newton's G - Casimir vacuum $\rightarrow \Lambda$ via T2418 - Gauge fields (for CMB structure formation) - Renormalization (for cosmological constant Λ derivation)

Inputs from Vol 3 (optional but useful): - Nuclear binding energies $\rightarrow$ BBN element abundances - Atomic structure $\rightarrow$ recombination history

Outputs to other Vol: - To Vol 9 Condensed Matter: gravity scale + cosmological context - To Vol 13 Information Theory / CSE meta: cosmological cycle as substrate-engineering framework - To Vol 14 Substrate Engineering Manual (if separate): SP-30 + cosmological signatures cross-anchor

Gap registry (what's missing for full Vol 4 v1.0)

- Ch 3 BST-SR / BST-GR boundary: when does substrate produce GR exactly? Multi-week theorem work
- Ch 9 Cosmological cycle Interstasis signatures: Task #267 just activated Friday; multi-month research
- Ch 12 SP-27 Observational Reanalysis Program: pending Casey scoping decision
- Quantum gravity bridge: how do D_{IV}^5 Bergman curvature $\rightarrow$ full Einstein equations? Multi-year per backlog
- Full GR-from-substrate derivation: T2418 partial; needs unified treatment

Status declarations

- v0.1 SCAFFOLD — INDEX + Architectural Scaffold filed, chapter outline drafted, cross-volume map drafted
- v0.3 chapter-grade narrative — when each chapter has chapter-grade content
- v0.4 reader-grade pedagogy — 3-level pedagogy applied per chapter
- v1.0 chapter-grade content REACHED — six-gate framework closed

Saturday 2026-05-23 EOD target: v0.1 SCAFFOLD COMPLETE; team can begin chapter content next.

Special note on Vol 4 signature status

Per CLAUDE.md Wednesday EOD framing: “D_IV⁵ as emergent TOE” — Casey: “I didn’t set out to build a theory of everything, yet D_IV⁵ seems to want to show us one.” Vol 4 GR & Cosmology is where this TOE-ness becomes operational for external readers. The cosmological constant, Hubble constant, CMB structure, BBN, dark energy and dark matter all derive from the same five integers as the Standard Model. Vol 4 is the signature volume that demonstrates BST’s cross-scale unification.

— Casey Koons + team, Vol 4 GR & Cosmology SCAFFOLD v0.1